

# Jaya Sharma

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## INTRODUCTION

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I'm a PhD student in Cybersecurity Engineering at the Technical University of Denmark (DTU), working under the supervision of Dr. Christian Majenz. My PhD research lies at the intersection of theoretical cryptography and quantum computing, with a focus on the security of cryptographic constructions in the quantum random oracle model. My other specific interests include zero-knowledge proofs, lattice-based cryptography, quantum information and error correction.

## EMPLOYMENT

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| <b>Technical University of Denmark (DTU)</b><br>PhD Student at Cybersecurity Engineering Section, DTU Compute<br><i>Supervisor:</i> Dr. Christian Majenz                                                                | 2024-27 |
| <b>Indraprastha Institute of Information Technology Delhi (IIITD)</b><br>PhD Junior Research Fellow in Department of Computer Science and Engineering<br><i>Area:</i> Lattice Based Cryptography and Quantum Computing. | 2022-24 |

## EDUCATION

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| <b>Indian Institute of Technology Roorkee (IITR)</b><br>Master of Science in Mathematics                         | 2020-22<br>Grade: 8.675/10 |
| <b>University of Delhi- Lady Shri Ram College For Women (LSR)</b><br>Bachelor of Science (Honors) in Mathematics | 2017-20<br>Grade: 9.081/10 |

## RESEARCH VISITS/INTERNSHIPS

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|                                                                                                                                                                                                                 |                 |
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| <b>Centrum Wiskunde &amp; Informatica (CWI)</b><br><i>Host:</i> Dr. Serge Fehr<br>-Working on security proofs for post-quantum non interactive zero knowledge protocols.                                        | 09/2026-01/2027 |
| <b>QWorld Association-QIntern</b> ( <i>Remote internship</i> )<br><i>Project:</i> Implementing Efficient block-encoding of dense matrices via QRAM.                                                             | 07-09/2025      |
| <b>Indian Institute of Science, Bangalore (IISc)</b><br><i>Mentor:</i> Dr. Chaya Ganesh<br>-Studied compression theory of sigma protocols and lattice based sigma protocols.                                    | 06-07/2023      |
| <b>Indian Institute of Science Education and Research (IISER) Bhopal</b> ( <i>Remote internship</i> )<br><i>Mentor:</i> Dr. Sanjay Singh<br><i>Project:</i> Introduction to Algebraic Function Fields and Codes | 06-08/2021      |

## PUBLICATIONS/PROJECTS

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| <b>Security of the Fischlin's Transform in Quantum Random Oracle Model</b><br><i>QCrypt'26, IACR ASIACRYPT'26</i><br>Christian Majenz, Jaya Sharma                                                                                                             | 2026 |
| <b>Implementating Efficient block-encoding of dense matrices via QRAM using the Qrisp framework</b><br><i>QWorld QIntern Project 2025 (Library currently in review by Qrisp Developers)</i><br>Viraj Dsouza, Kalpesh Bhatnagar, Kyle Carlo Lasala, Jaya Sharma | 2025 |
| <b>A Survey on Quantum Error Correcting Codes</b>                                                                                                                                                                                                              | 2023 |
| <b>Elliptic Curves Over Complex Numbers</b><br><i>Master's Thesis</i><br>Supervisor- Dr. A. Swaminathan                                                                                                                                                        | 2022 |
| <b>Brent-Dekker Method: An Attempt to Decode</b><br><i>Eclat- A Mathematics Journal, Vol 11</i><br>Dr. Monika Singh, Shagun Agrawal, Jaya Sharma                                                                                                               | 2020 |

## PRESENTATIONS

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|---------------------------------------------------------------------------------------------|---------|
| Security of the Fischlin Transform in Quantum Random Oracle Model- QCrypt UOttawa           | 08/2026 |
| Fiat Shamir Transform in QROM- Crypto Reading Group DTU                                     | 10/2024 |
| Lattice Based Sigma Protocols- Theory Reading Group IIITD                                   | 10/2023 |
| Trapdoors for Hard Lattices and New Cryptographic Constructions-Theory Reading Group IIITD  | 05/2023 |
| The Rabin Cryptosystem: Mathematical Foundations and Implementation- IIITD                  | 12/2022 |
| Coherent Measures of Financial Risk- MSc Mathematics Seminar, IITR                          | 10/2021 |
| Elliptic Curve Cryptography- Department of Mathematics, LSR                                 | 05/2019 |
| Latin Squares- Recreation, Disproving Euler to Cryptography- Department of Mathematics, LSR | 05/2018 |
| Applications of Congruence Modulo- Department of Mathematics LSR                            | 02/2018 |

## TEACHING ASSISTANCE

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|------------------------------------------|-----------|
| Discrete Mathematics (DTU)               | 2024,2025 |
| Graph Theory (DTU)                       | 2025      |
| Applied Cryptography (DTU)               | 2024      |
| Introduction To Cryptography (IIT Delhi) | 2023      |
| Discrete Mathematics (IIITD)             | 2022      |

## AWARDS

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| Thomas B. Thriges Fond Grant, William Demant Fonden Grant, Otto Mønstedts Fond Grant<br>-Danish Grants to fund research stay at CWI Amsterdam.                                                                                                                         | 2026 |
| ACM-India Anveshan Setu Fellowship<br>- Awarded by Association for Computing Machinery(ACM) India to work at IISc.                                                                                                                                                     | 2023 |
| Joint CSIR-UGC NET (JRF) Fellowship Mathematical Sciences<br>-Awarded PhD Research Fellowship for securing All India Rank 122/30942 candidates in the exam conducted jointly by Council for Scientific and Industrial Research and University Grants Commission India. | 2022 |
| IIT JAM Mathematics: secured All India Rank 143/14374 candidates<br>- Joint Admission Test for Master of Science Courses at Centrally Funded Technical Institutions in India.                                                                                          | 2020 |

## PROGRAMMING

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C++, [Python](#), Qiskit.

## WORKSHOPS/SUMMER SCHOOLS/CONFERENCES

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|------------------------------------------------------------------------------------|---------|
| Qcrypt- Ottawa, Canada                                                             | 08/2026 |
| Eurocrypt- Rome, Italy                                                             | 05/2026 |
| DQC Scientific Quantum Conference 2026- Aalborg University, Denmark                | 04/2026 |
| Theory of Cryptography Conference (TCC) 2025-Aarhus, Denmark                       | 12/2025 |
| Workshop on Mathematics of Post-Quantum Cryptography- University of Zurich         | 06/2025 |
| Quantum Safe Internet (QSI) Workshop- Technical University of Denmark, Lyngby      | 05/2025 |
| IACR Summer School on Post Quantum Cryptography- University of Warsaw              | 07/2024 |
| ACM India Grad Cohort- IISc Bangalore                                              | 07/2023 |
| CSE Research Symposium- IIT Bombay                                                 | 03/2023 |
| Vigyan Vidushi Mathematics Summer School- TIFR Mumbai                              | 07/2021 |
| STCS Vigyan Vidushi Summer School on Theoretical Computer Science- TIFR Mumbai     | 06/2021 |
| Virtual Math Fest India                                                            | 2020    |
| Workshop: Calculus the Geometrical Way- Miranda House College, University of Delhi | 09/2019 |

## OTHER ACTIVITIES/SERVICE

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| Sub-Reviewer Eurocrypt'27                                                     | 2026 |
| PhD Committee Member- DTU Compute                                             | 2025 |
| Illustrator- LSR College Magazine                                             | 2019 |
| Event Coordinator- Enigma, Annual Academic Meet Department of Mathematics LSR | 2019 |

## OTHER INFORMATION

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**Citizenship-** Indian ; **Current Visas-** Denmark Residence Permit/ Schengen- till 2030, Canada- till 2032  
**Languages-** English, Danish (elementary Proficiency), Hindi (mother tongue)